

DAILY JOURNALS

Date of Journal	June 22, 2026
Time of Duty	9:00:00 AM-11:30:00 AM, 12:00:00 PM-5:00:00 PM
No. of Hours	7:30:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
1. Conducted orientation as an intern for the National Institute of Physics. 2. Introduced to Paper by Ramon Bagunu and Eric Galapon. 3. Read paper and rederive concepts.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers.	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? I had difficulty understanding the paper that was given to me while at the same time being accustomed to the environment I am exposed with. To solve this, I researched seminal papers by which the main paper was based on.		
What are your tasks tomorrow? To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	June 23, 2026
Time of Duty	8:04:00 AM-5:50:00 PM
No. of Hours	9:46:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
1. Conducted a primary reading of the paper by Bagunu and Galapon. 2. Develop a thorough understanding of the results of the paper. 3. Write down/encode my understanding of the paper.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers.	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Understanding the paper given, but was able to respond to it by researching further and finding other avenues to educate myself.		
What are your tasks tomorrow? To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	June 24, 2026
Time of Duty	8:15:00 AM-8:00:00 PM
No. of Hours	11:45:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
4. Started to accomplish rederivations of the start of the paper. Investigated images of Weyl, simplest symmetrization, and Born-Jordan ordering rules. 5. Provide a preliminary report to our supervisor on progress to the paper studied. Refer to the appendix section OUTPUT for progress (Section 1). 6. Continue reading and writing for the results found in the paper.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Report to the supervisor on the current progress on learning the paper.
What challenges did you encounter and how were you able to respond to it?		
Nerves with reporting, was unable to clearly report my progress due to anxiety. However, I was still able to relay a significant portion of what I have learned so far.		
What are your tasks tomorrow?		
To further read and increase progress and understanding the paper given.		

DAILY JOURNALS

Date of Journal	June 25, 2026
Time of Duty	9:05:00 AM-5:50:00 PM
No. of Hours	8:45:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
7. Read upon research regarding quantization. Investigated the relationship of the Schrodinger and Heisenberg pictures of quantum mechanics. 8. Rederive formulas and results thereof. 9. Write on scratch notebook as recommended by adviser.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Understanding the paper given, but was able to respond to it by researching further and finding other avenues to educate myself.		
What are your tasks tomorrow? To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	June 26, 2026
Time of Duty	8:15:00 AM-1:15:00 PM, 4:00:00 PM-7:00:00 PM
No. of Hours	8:00:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
10. Research on the resulting differential quotient from the Born-Jordan quantization technique and ordering rule. Refer to the appendix section OUTPUT for progress (Section 2). 11. Rederive formulas and results thereof. 12. Write on scratch notebook as recommended by adviser.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it?		
Understanding the paper given, but was able to respond to it by researching further and finding other avenues to educate myself.		
What are your tasks tomorrow?		
To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	June 29, 2026
Time of Duty	8:20:00 AM-5:50:00 PM
No. of Hours	9:30:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
13. Continuation of the derivation of the differential quotient resulting in Born-Jordan images. Refer to the appendix section OUTPUT for progress (Section 3). 14. Rederive formulas and results thereof. 15. Write on scratch notebook as recommended by adviser.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it?		
Messy handwriting, eventually switched the digital media for notes for more organization.		
What are your tasks tomorrow?		
To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	June 30, 2026
Time of Duty	8:10:00 AM-2:10:00 PM, 3:00:00 PM-8:00:00 PM
No. of Hours	11:00:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
16. Derivation of the images as a result of the modifications to the differential quotient of first type. Refer to the appendix section OUTPUT for progress (Section 4). 17. Rederive formulas and results thereof. 18. Write on scratch notebook as recommended by adviser.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it?		
Understanding how such constants added to the original formula were added (what was the authors' basis for adding them). Consulted with supervisor for help.		
What are your tasks tomorrow?		
To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 1, 2026
Time of Duty	8:51:00 AM-8:00:00 PM
No. of Hours	11:09:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
19. Reading and understanding of Suzuki's "General Formulation of Quantum Mechanics." Refer to the appendix section OUTPUT for progress (Section 5). 20. Start of derivation of the theorems. 21. Rederive formulas and results thereof.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Understanding Banach spaces and the basis for doing the mathematical methods involved. Fixed by researching books further on Banach spaces and group theory.		
What are your tasks tomorrow? To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 2, 2026
Time of Duty	7:35:00 AM-5:30:00 PM
No. of Hours	9:55:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
22. Start of derivation of the theorems. 23. Rederive formulas and results thereof. 24. Write into notebook/TeX file on progress. Refer to the appendix section OUTPUT for progress (Section 5).		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Understanding why such methods could be applied to operators. However, consulting books around the office helped me understand it.		
What are your tasks tomorrow? To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 3, 2026
Time of Duty	7:45:00 AM-12:40:00 PM, 1:35:00 PM-6:00:00 PM
No. of Hours	9:20:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
25. Continuation of derivation of the theorems. 26. Researched on the derivation of Euler's reflection formula for negative factorials. Rederive formulas and results thereof. 27. Write into notebook/TeX file on progress. Refer to the appendix section OUTPUT for progress (Section 5).		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it?		
Understanding what Euler's reflection formula does and how it was derived. Big thank you to the resources and books which helped!		
What are your tasks tomorrow?		
To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 7, 2026
Time of Duty	9:13:00 AM-3:19:00 PM
No. of Hours	6:06:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
28. Continuation of derivation of the theorems. 29. Finished the derivation of the theorems. 30. Write into notebook/TeX file on progress. Refer to the appendix section OUTPUT for progress (Section 5).		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Anxiety with presenting. Kept my composure and assured myself that I know what I am going to be reporting.		
What are your tasks tomorrow? To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 9, 2026
Time of Duty	9:00:00 AM-4:00:00 PM, 5:50:00 PM-7:40:00 PM
No. of Hours	8:50:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
31. Initial report of the progress to another professor in the Theoretical Physics Group, Dr. Ralph Farrales, who actually graduated from UP Manila! 32. Incorporated comments and suggestions by Dr. Farrales. 33. Write into notebook/TeX file on progress. Refer to the appendix section OUTPUT for progress (Section 5).		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it?		
Anxiety with presenting. Kept my composure and assured myself that I know what I am going to be reporting.		
What are your tasks tomorrow?		
To further read and ready my report for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 13, 2026
Time of Duty	8:40:00 AM-3:40:00 PM, 5:00:00 PM-8:20:00 PM
No. of Hours	10:20:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
34. Deviated a bit from the paper assigned. 35. I started to read the unpublished dissertation of Dr. Herbert Domingo. 36. Write into notebook/TeX file on progress.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Intensive read considering the evident lack of knowledge on the topic. Explored and fixed by further reading and referencing authors.		
What are your tasks tomorrow? To further read and ready my report to Dr. Eric Galapon for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 14, 2026
Time of Duty	8:30:00 AM-11:30:00 AM, 12:10:00 PM-3:20:00 PM
No. of Hours	6:10:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
37. Deviated a bit from the paper assigned. 38. Further started to read the unpublished dissertation of Dr. Herbert Domingo. 39. Write into notebook/TeX file on progress.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Intensive read considering the evident lack of knowledge on the topic. Explored and fixed by further reading and referencing authors.		
What are your tasks tomorrow? To further read and ready my report to Dr. Eric Galapon for progress regarding the paper.		

DAILY JOURNALS

Date of Journal	July 15, 2026
Time of Duty	8:50:00 AM-3:30:00 PM
No. of Hours	6:40:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
40. Returned to the paper assigned. Explored and finished the section regarding the actual quantization of the Hamilton Equations of Motion. 41. Finished the analysis on example of the harmonic oscillator. 42. Write into notebook/TeX file on progress. Refer to the appendix section OUTPUT for progress (Section 6 and Appendix).		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it?		
Logic heavy considering its proving. Consulted with other papers to understand the results proper.		
What are your tasks tomorrow?		
To further read and ready my report for progress regarding the paper on Friday.		

DAILY JOURNALS

Date of Journal	July 16, 2026
Time of Duty	8:50:00 AM-4:00:00 PM
No. of Hours	7:10:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
43. Completion of the analysis of the paper. 44. Refined the contents of the report. 45. Write into notebook/TeX file on progress.		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it?		
Lots of typos identified. Fixed by reviewing and rechecking results with own calculations.		
What are your tasks tomorrow?		
To further read and ready my report for progress regarding the paper on Friday.		

DAILY JOURNALS

Date of Journal	July 16, 2026
Time of Duty	8:50:00 AM-5:00:00 PM
No. of Hours	8:10:00

Which of the following PLOs were met for today's training? Put a check (/) on the corresponding objective.	
/	Apply theoretical, computational, and experimental physics in a workplace;
NA	Conduct research related to health and medical physics that contribute to national development;
NA	Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
NA	Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
/	Synthesize and effectively communicate scientific information in both English and Filipino;and
/	Act in recognition of professional, social, and ethical responsibility.

What were the tasks you accomplished today? Cite at least three.		
46. Completion of the analysis of the paper. 47. Reported to Dr. Galapon the finished paper analysis. 48. Write into notebook/TeX file on progress. Refer to OUTPUT for whole analysis./		
How did you spend today's training? Create a flow chart.		
Read the paper given and try to take notes, rederive if possible.	Try to understand. If not, research the topic with other seminal papers	Return to the original paper and re-read with a new perspective.
What challenges did you encounter and how were you able to respond to it? Presentation was cut short due to time constraints of Dr. Galapon. Made up by allotting more hours for reporting the following week.		
What are your tasks tomorrow? To explore other quantization methods and conduct the same calculations as done since the first day.		